

Assignment

1. Write an iterative function to compute the factorial of a natural number

2. Write a function to compute $\frac{x^n}{n!}$ given a float x and natural number n as arguments

3. Write a function to iteratively sum up the value $\frac{x^n}{n!}$ from $n = 1$ to a given N for a given x , i.e.,

$$F(x, N) = 1 + \sum_{i=1}^N \frac{x^i}{i!}$$

4. Write a function to iteratively sum up the value $\frac{x^n}{n!}$ from $n = 1$ to a chosen value of N' such that $F(x, N') - F(x, N' - 1) < \epsilon$ for a given real number x and positive small number ϵ

5. Choose two real numbers p and q and compute the following two values

$$v_1 = F(p, 100) * F(q, 100)$$

$$v_2 = F(p + q, 100)$$

Compute the difference $v_1 - v_2$ and comment on what you see.